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STA416

CHAPTER 5

Continuous Random Variables and Probability Distributions

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Continuous Random Variable

- A random variable is called continuous if it can assume all possible values in the possible range of the random variable where the data can take infinitely many values.
- Some examples of the continuous random variables are:
 - The computer time (in seconds) required to process a certain program.
 - The time for a baby to gain the weight of 5.5kg.
 - The amount of rain falls in the certain city.
 - The heat gained by a ceiling fan when it has worked for one hour.

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Probability Density Function

- The probability function of the continuous RV is called the probability density function (p.d.f).
- The number of possible outcomes of a continuous RV is uncountable infinite.
- This means that we must calculate a probability for a continuous RV over an interval and not for any particular point.
- This probability can be interpreted as an area under the graph between the interval from a to b.

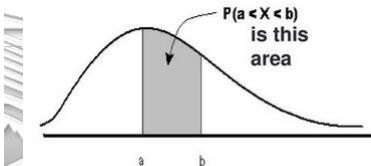
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Probability Density Function

- How to find the probability of some interval of the continuous RV?

Use any one of these two methods:

- ✓ Integral calculus
- ✓ Area by geometrical diagrams (this method is easy to apply when is a simple linear function).



- $P(a \leq X \leq b)$ is a given by the area under

- We can't find probabilities for exact outcomes.
- For example: $P(X = 2) = 0$. (Also only range ni)
- Instead we can find probabilities for a range of values.

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Definition of Probability Density Function

The function $f(x)$ is a probability density function (pdf) for continuous RV X , defined over the set of real numbers, if

1. $f(x) \geq 0$, for all $x \in \mathbb{R}$
 2. $\int_{-\infty}^{\infty} f(x) dx = 1$
 3. $P(a < X < b) = \int_a^b f(x) dx$
- } if not, ia bukan continuous RV.

Note:

- it is a function.
1. $f(x)$ is **NOT a probability**. To find the probability that X lies between two values a and b , say where $a \leq b$, we use Property 3 above.
 2. For any single value c , $P(X=c)=0$ (any exact num must equal 0)
 3. For any two numbers a and b with $a < b$,

$$\begin{aligned} P(a \leq X \leq b) &= P(a \leq X < b) \\ &= P(a < X \leq b) \\ &= P(a < X < b) \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{ sama je semua}$$

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Example:

A continuous r.v X has a pdf:

$$f(x) = \begin{cases} \frac{x^2}{3} & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$f(x) = \begin{cases} \frac{x^2}{3} & x = -1, 0, 1, 2 \\ 0 & \text{elsewhere} \end{cases}$$

→ not in interval so this is Chapter 4 bkn continuous RV.

(a) Verify that $f(x)$ is a density function.

(b) Find $P(0 < X \leq 1)$.

$$\begin{aligned} \text{a) } \int_{-\infty}^{\infty} f(x) dx &= \int_{-1}^2 \frac{x^2}{3} dx + \int_{-\infty}^{-1} 0 dx + \int_2^{\infty} 0 dx \\ &= \int_{-1}^2 \frac{x^2}{3} dx \\ &= \frac{1}{3} \left[\frac{x^3}{3} \right]_{-1}^2 \\ &= \frac{1}{3} \left[\frac{8}{3} - \left(-\frac{1}{3} \right) \right] = \frac{1}{3} \left[\frac{9}{3} \right] = 1 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{b) } \int_0^1 \frac{x^2}{3} dx &= \left[\frac{x^3}{9} \right]_0^1 \\ &= \left[\frac{1^3}{9} - \frac{0^3}{9} \right] \\ &= \frac{1}{9} \quad \checkmark \end{aligned}$$

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Example:

A continuous r.v X has a pdf:

$$f(x) = \begin{cases} kx & 0 \leq x \leq 1 \\ k(2-x) & 1 < x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

- 1) Find k
- 2) $P(0.5 < x \leq 1.5)$

$$\begin{aligned} 1) & k \int_0^1 x \, dx + k \int_1^2 (2-x) \, dx = 1 \\ & k \left[\frac{x^2}{2} \right]_0^1 + k \left[2x - \frac{x^2}{2} \right]_1^2 = 1 \\ & k \left[\frac{1^2}{2} - \frac{0^2}{2} \right] + k \left[\left(2(2) - \frac{2^2}{2} \right) - \left(2(1) - \frac{1^2}{2} \right) \right] \\ & \frac{1}{2}k + \frac{1}{2}k = 1 \\ & k = 1 \quad * \end{aligned}$$

$$\begin{aligned} 2) & \int_{0.5}^1 x \, dx + \int_1^{1.5} (2-x) \, dx \\ & = \left[\frac{x^2}{2} \right]_{0.5}^1 + \left[2x - \frac{x^2}{2} \right]_1^{1.5} \\ & = \left[\frac{1^2}{2} - \frac{0.5^2}{2} \right] + \left[\left(2(1.5) - \frac{1.5^2}{2} \right) - \left(2(1) - \frac{1^2}{2} \right) \right] \\ & = \frac{3}{8} + \left(1 - \frac{5}{8} \right) \\ & = \frac{3}{4} \quad * \end{aligned}$$

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Cumulative distribution Function for Continuous Random Variable

**Jawab dalam exam.*

The cumulative distribution function $F(x)$ of a continuous random variable X with density function $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) \, dt, \quad \text{for } -\infty < x < \infty.$$

For each x , $F(x)$ is the area under the density curve to the left of x .

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Some important formulae for cumulative distribution function (cdf) of a continuous r.v.

1. $P(a \leq X \leq b) = F(b) - F(a)$

The above formula is also applicable to:

- a) $P(a \leq X < b)$
- b) $P(a < X \leq b)$
- c) $P(a < X < b)$

2. It follows the fundamental theorem of calculus that if we are given the pdf by differentiating

$$F'(x) = f(x) = \frac{dF(x)}{dx}$$

provided the derivative exist.

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Example:

A continuous r.v X has a pdf:

$$f(x) = \begin{cases} \frac{x^2}{3} & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$\int_{-1}^2 x^2 dx = 0$$

- a) Find $F(x)$
- b) By using $F(x)$ above, evaluate $P(0 < X \leq 1)$

Solution:

a) $F(x) = \int_{-\infty}^x f(t) dt$

b)

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Mean (Expected Value) of Continuous Random Variable

Let X be a continuous r.v. with probability distribution $f(x)$.

1. The mean or expected value of X is

$$\mu = E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

2. The mean or expected value of the r.v. $g(X)$ is

$$\mu_{g(x)} = E[g(x)] = \int_{-\infty}^{\infty} g(x) \cdot f(x) dx$$

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Variance of Continuous Random Variable

Let X be a continuous r.v. with probability distribution $f(x)$.

3. The variance of X is

$$V(X) = \sigma^2 = E(X^2) - [E(X)]^2$$

Where

$$\mu = E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$\mu^2 = E(X^2) = \int_{-\infty}^{\infty} x^2 \cdot f(x) dx$$

Means and Variances of Linear Combination of Continuous RV are same with the discrete case in Chapter 3.4

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Example :

The weekly demand for a drinking-water product, in thousands of liters, from a local chain of efficiency stores is a continuous random variable X having the probability density

$$f(x) = \begin{cases} 2(x-1) & 1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{Variance} \\ E(X^2) - [E(X)]^2$$

Find the mean and variance of X .

$$\begin{aligned} E(X) &= \int_1^2 x f(x) dx = \int_1^2 x \cdot 2(x-1) dx \\ &= \int_1^2 2x^2 - 2x dx = \left[\frac{2x^3}{3} - \frac{2x^2}{2} \right]_1^2 \\ &= \left[\frac{2(8)}{3} - \frac{2(4)}{2} \right] = \frac{8}{3} - 4 = -\frac{4}{3} \end{aligned}$$

$$E(X^2) = \int_1^2 x^2 \cdot 2(x-1) dx = \int_1^2 2x^3 - 2x^2 dx = \left[\frac{2x^4}{4} - \frac{2x^3}{3} \right]_1^2 = \left[\frac{x^4}{2} - \frac{2x^3}{3} \right]_1^2 = \left[\frac{16}{2} - \frac{16}{3} \right] - \left[\frac{1}{2} - \frac{2}{3} \right] = \frac{8}{3} - \frac{1}{6} = \frac{15}{6} = \frac{5}{2}$$

$$V(X) = E(X^2) - [E(X)]^2 = \frac{5}{2} - \left(-\frac{4}{3}\right)^2 = \frac{5}{2} - \frac{16}{9} = \frac{45}{18} - \frac{32}{18} = \frac{13}{18}$$

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Example

Let X be a random variable with density function

$$f(x) = \begin{cases} \frac{x^2}{3} & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

$E(X)$

Find the expected value of $g(X) = 4X + 3$.

Solution:

$$\begin{aligned} E(g(X)) &= E(4X + 3) \\ &= \int_{-1}^2 (4x + 3) \left(\frac{x^2}{3}\right) dx \\ &= \int_{-1}^2 \left(\frac{4x^3}{3} + \frac{3x^2}{3}\right) dx \\ &= \left[\frac{4x^4}{12} + \frac{3x^3}{3} \right]_{-1}^2 \\ &= \left[\frac{x^4}{3} + x^3 \right]_{-1}^2 \\ &= \left(\frac{16}{3} + 8 \right) - \left(\frac{1}{3} - 1 \right) \\ &= \frac{16}{3} + 8 - \frac{1}{3} + 1 \\ &= \frac{15}{3} + 9 = 5 + 9 = 14 \end{aligned}$$

$$\begin{aligned} E(4X + 3) &= 4E(X) + 3 \\ &= E(X) = \int_{-1}^2 \frac{x^2}{3} dx \\ &= \left[\frac{x^3}{9} \right]_{-1}^2 \\ &= \frac{8}{9} - \left(-\frac{1}{9}\right) = \frac{9}{9} = 1 \\ &= 4(1) + 3 = 7 \end{aligned}$$

You also can solve this problem by using method of Linear Combination of Continuous RV.

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Exercise 1:

A continuous random variable X has the following probability density function.

$$f(x) = \begin{cases} kx(3-x) & , 0 \leq x \leq 3 \\ 0 & , \text{otherwise} \end{cases}$$

where k is a constant.

- Show that $k = \frac{2}{9}$.
(3 marks)
- Find $P\left(\frac{3}{4} < X < \frac{5}{2}\right)$.
(3 marks)
- Calculate $E(X)$ and $\text{Var}(X)$.
(7 marks)
- Find $\text{Var}(2X - 1)$.
(2 marks)

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Exercise 2

Let X be a continuous random variable with the following probability density function.

$$f(x) = \begin{cases} \frac{1}{4}x & ; 0 \leq x < 2 \\ \frac{1}{4}(4-x) & ; 2 \leq x < 4 \\ 0 & ; \text{otherwise} \end{cases}$$

Calculate

- $P(1 \leq X \leq 3)$.
(5 marks)
- $E(X)$.
(4 marks)
- $\text{Var}(X)$.
(6 marks)

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